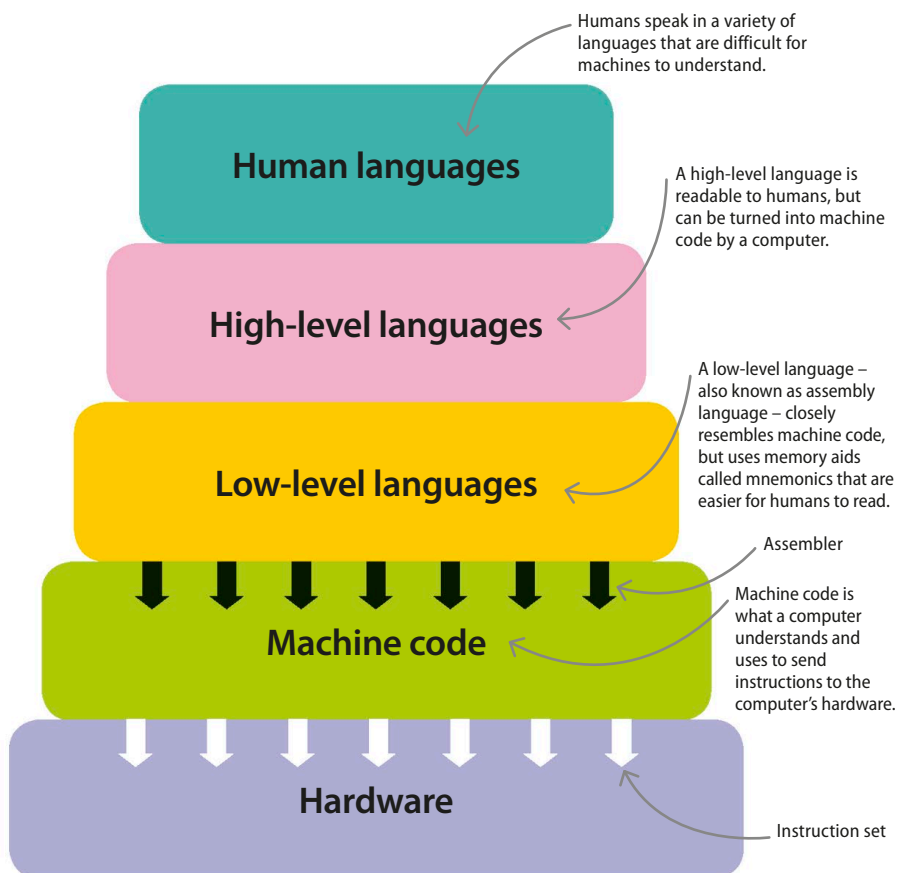


Translation

Translation is the process of converting one programming language into another. It makes coding more human-friendly by breaking down a high-level language into a low-level one.

High- and low-level languages

A manager of a clothing company doesn't need to know how many socks are delivered to each store. Instead, they are required to take a high-level view and only make decisions that are critical to the company's future. In a similar way, high-level programming languages don't worry about details. Memory management, converting instructions into bits, and varying voltage within physical circuits are all details hidden away from modern programmers. Instead, these are automatically handled by special programs called translators. This makes it easier for humans to focus on logic and complex algorithms, and the computer to handle everything else.



SEE ALSO

Assemblers, interpreters, and compilers	112–113 >
What do programming languages do?	118–119 >
C and C++	126–127 >
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"C is quirky, flawed, and an enormous success."

Dennis Ritchie (1941–2011), American creator of the C programming language

IN DEPTH

Pros and cons

High-level languages may seem a better option, but they fall short in certain scenarios.

Pros

- Easier to learn.
- Easier to read and write. Difficult algorithms don't have to be converted into abstract strings of 0s and 1s.
- Easier to understand. Code can be shared, debugged, and augmented more quickly.
- Great for big, collaborative projects.

Cons

- Programs are bigger and slower. Since developers can't access the nuts and bolts of the code, programs are difficult to streamline.
- Security risks – developers can't check if memory is properly secured and wiped.
- The popularity of high-level languages means that many computer scientists don't fully understand operating systems and hardware. This can lead to flaws in software design.